Q4 and Q8 accuracy in Yoruba and English

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Exploratory analysis. Not preregistered or peer reviewed.

Research question

Does the change in AfriXNLI accuracy from Q8_0 to Q4_K_M differ between Yoruba and English for Qwen2.5-1.5B-Instruct?

I compare two published GGUF versions of Qwen2.5-1.5B-Instruct on AfriXNLI. I chose this narrower question after examining the wider benchmark results. This note uses existing answers. I have not collected new model responses for it.

Input language	Q8_0	Q4_K_M
English	300 questions	Same 300 questions
Yoruba	300 parallel translations	Same 300 translations

Method

For each language, I subtract Q8 accuracy from Q4 accuracy. I then subtract the English change from the Yoruba change. A negative result would mean a worse change for Yoruba. Each question has four related outcomes, so the analysis keeps 300 question pairs together.

Both model files come from the same version of the official GGUF release. The records specify the same workspace, runtime binary, two-thread CPU setup, 2,048-token context, English instructions and labels, greedy decoding, seed, 128-token output limit and answer scorer. I also check that request sizes and prompt-token counts agree between configurations for each item.

I parse and score the selected answers again, check them against the stored results and retain request failures. I reject missing outcomes and repeated successful responses. The uncertainty estimates use 10,000 bootstrap samples, seed 20261102, with all four outcomes for a question sampled together. I report 95% percentile intervals for these recorded runs.

Limits of the existing runs

The configurations ran on different dates on a shared host, and their order was not randomised. I do not have the full history of the original weights, conversion and calibration. Q8_0 is also compressed, and the recipes differ as well as nominal precision. These records cannot isolate the effect of bit width.

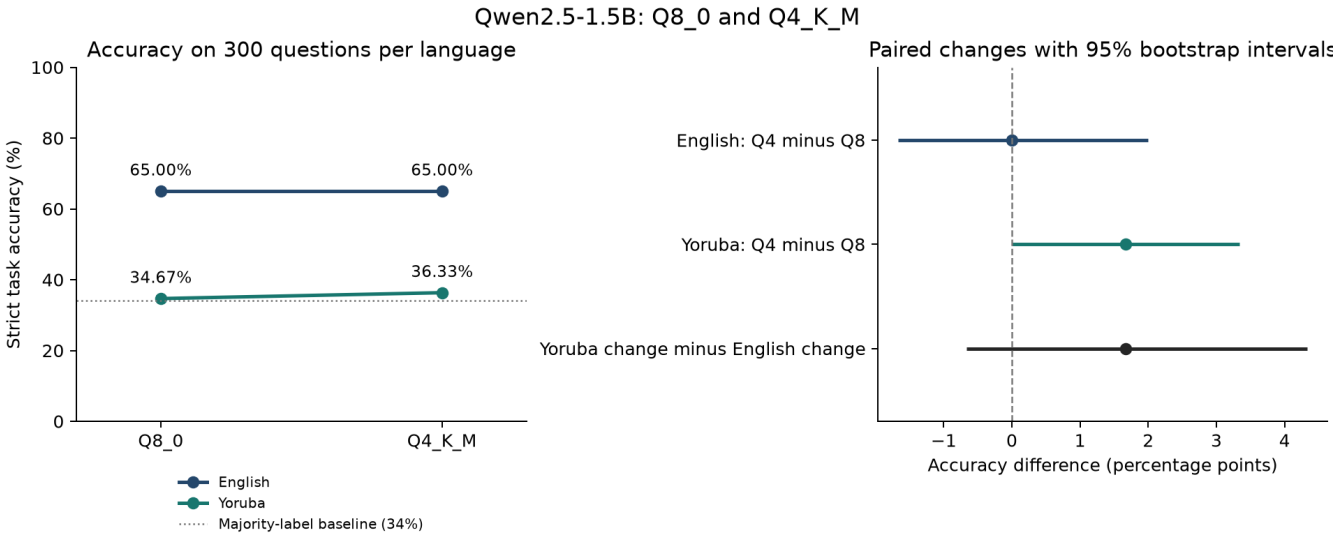
Other tasks, languages, models, providers, translation pipelines and timing repeats are outside this analysis. I do not use the results to draw conclusions about serving cost, latency, phones or production reliability.

Accuracy and paired differences

Main comparison: +1.67 pp [-0.67, +4.33]. The 95% bootstrap interval includes zero. I cannot conclude that Yoruba had a greater compression penalty. This does not show that the configurations are equivalent or establish a general effect of bit width.

Language	Configuration	Correct / 300	Accuracy
English	Q8_0	195	65.00%
English	Q4_K_M	195	65.00%
Yoruba	Q8_0	104	34.67%
Yoruba	Q4_K_M	109	36.33%

Paired contrast	Estimate [95% bootstrap interval]
English: Q4 minus Q8	+0.00 pp [-1.67, +2.00]
Yoruba: Q4 minus Q8	+1.67 pp [+0.00, +3.33]
Yoruba change minus English change	+1.67 pp [-0.67, +4.33]



The English configurations have the same overall accuracy, but four questions are answered correctly only by Q4 and four only by Q8. The corresponding Yoruba counts are six and one. The bootstrap keeps the related answers together when sampling questions.

The scorer can read all 1,200 final responses, and none stops at the output token limit. One English Q8 request failed before a successful retry. That attempt remains in the source log. Answer format and output limits therefore do not explain these accuracy values.

Errors and remaining checks

Neutral-class results

Language	Configuration	Neutral recall
English	Q8_0	0.98%
English	Q4_K_M	4.90%
Yoruba	Q8_0	0.00%
Yoruba	Q4_K_M	0.00%

Neither Yoruba configuration predicts neutral. English neutral recall is poor too. Each language has 99 entailment, 102 neutral and 99 contradiction reference labels. Always choosing the most common label would score 34% on this sample. Most Yoruba predictions are entailment. I have not established whether the cause is model capability, prompting, translations or something else.

Reported prompt lengths are 72 to 124 tokens in English and 92 to 281 in Yoruba. Instructions and labels are in English; the premise and hypothesis are in the assigned language. I kept the original prompt and answer parser.

Before a confirmation run

The intervals describe variation across questions in the existing runs. They do not cover changes in host activity, differences between repeated runs, translation errors or the decision to narrow the question after seeing results. I also need a fuller record of the starting weights and conversion steps. These benchmark translations have not received an independent native-speaker audit in this study. The agent-safety volunteers reviewed a different dataset.

I have listed the other 300 parallel questions from the same 600-row benchmark. They do not overlap with this analysis, and I have not run confirmation inference on them. Before doing so, I would agree the block order and repeated subset, use a reserved host, record how both model files were produced, balance which configuration runs first and retain every failed request. Those execution details are not finalised. The questions come from the same benchmark, rather than a separate corpus.

Questions for review

Does this comparison address a useful question? Given the poor class behaviour, should I first establish a suitable model and prompt in a separate development phase before running confirmation?

Data and code

Protocol, results, source hashes and analysis code.

From the repository root, install the dev and review extras, then run python -m bench.data, python -m bench.focused_nli, python -m bench.focused_review and pytest -q. Data preparation reads the pinned benchmark. The analysis sends no requests to a model. The original run files and earlier protocol are unchanged.